

00_Switch Configuration

Cisco IOS Switch Configuration Task

The more you practice learning how to configure a switch in different ways, the easier it will become to memorize the commands and task given to you. Some of the objectives you will have to complete will not be used in a real working environment but the key to understanding why something **is/isn't** used is to understand how it works.

Unlike your walkthrough Exercises These tasks will assume you have a basic understanding of the task given. For example unless stated otherwise used the most convenient form of connection to a device, whether that be Serial-to-USB, SSH or even Telnet.

What you will need

- A LAN cable
- Serial-to-USB cable
- 2 PCs or Laptops
- A Cisco Switch with at *least* 6 ports (for this example we'll use a [Cisco IE 3000](#))
- PuTTY or MobaXterm

Complete the following:

1. Change the name of the Switch to **IND-ED-TEST**
2. Create VLANs, assign 1 port to each VLAN and give them their own names (One of them has to be named **MGMT**) - 10, 20, 30, 40, 50
3. Give the **MGMT (MANAGEMENT)** VLAN an IP address - **192.168.10.2/24**
4. Enable SSH and Telnet

Documentation

Step 1: Change the name of the Switch to **IND-ED-TEST**

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname IND-ED-TEST
IND-ED-TEST(config)#
```

Step 2: Create VLANs, assign 1 port to each and name them

First let's see what we're working here bub.

```
IND-ED-TEST(config)#do show vlan
```

VLAN	Name	Status	Ports
1	default	active	Fa1/1, Fa1/2, Fa1/3, Fa1/4 Gi1/1, Gi1/2
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

1. Now let's Create them

```
IND-ED-TEST(config-vlan)#vlan 50
IND-ED-TEST(config-vlan)#vlan 40
IND-ED-TEST(config-vlan)#vlan 30
IND-ED-TEST(config-vlan)#vlan 20
IND-ED-TEST(config-vlan)#vlan 10
```

Cool beans, how's it looking, are we cooking good looking?

```
IND-ED-TEST(config)#do show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa1/1, Fa1/2, Fa1/3, Fa1/4 Gi1/1, Gi1/2
10	VLAN0010	active	
20	VLAN0020	active	
30	VLAN0030	active	
40	VLAN0040	active	
50	VLAN0050	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

2. Now let's name them twin

```

IND-ED-TEST(config-vlan)#vlan 10
IND-ED-TEST(config-vlan)#name USERS
IND-ED-TEST(config-vlan)#vlan 20
IND-ED-TEST(config-vlan)#name VOICE
IND-ED-TEST(config-vlan)#vlan 30
IND-ED-TEST(config-vlan)#name MGMT
IND-ED-TEST(config-vlan)#vlan 40
IND-ED-TEST(config-vlan)#name IOT
IND-ED-TEST(config-vlan)#vlan 50
IND-ED-TEST(config-vlan)#name GUEST

```

**Is it coming or are you :)

```

```cisco
IND-ED-TEST(config)#do show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
Vlan1	unassigned	YES	unset	administratively down	down
FastEthernet1/1	unassigned	YES	unset	down	down
FastEthernet1/2	unassigned	YES	unset	down	down
FastEthernet1/3	unassigned	YES	unset	down	down
FastEthernet1/4	unassigned	YES	unset	down	down
GigabitEthernet1/1	unassigned	YES	unset	down	down
GigabitEthernet1/2	unassigned	YES	unset	down	down

```

IND-ED-TEST(config)#do show vlan brief

```

VLAN	Name	Status	Ports
1	default	active	Fa1/1, Fa1/2, Fa1/3, Fa1/4 Gi1/1, Gi1/2
10	USERS	active	
20	VOICE	active	
30	MGMT	active	
40	IOT	active	
50	GUEST	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

### 3. Assign 1 port to each VLAN

```

IND-ED-TEST(config)#
IND-ED-TEST(config)#interface FastEthernet1/1
IND-ED-TEST(config-if)#switchport mode access
IND-ED-TEST(config-if)#switchport access vlan 1
IND-ED-TEST(config-if)#exit
IND-ED-TEST(config)#interface FastEthernet1/2
IND-ED-TEST(config-if)#switchport mode access
IND-ED-TEST(config-if)#switchport access vlan 10
IND-ED-TEST(config-if)#exit
IND-ED-TEST(config)#interface FastEthernet1/3
IND-ED-TEST(config-if)#switchport mode access
IND-ED-TEST(config-if)#switchport access vlan 20
IND-ED-TEST(config-if)#exit
IND-ED-TEST(config)#interface FastEthernet1/4
IND-ED-TEST(config-if)#switchport mode access
IND-ED-TEST(config-if)#switchport access vlan 30
IND-ED-TEST(config-if)#exit
IND-ED-TEST(config)#interface GigabitEthernet1/1
IND-ED-TEST(config-if)#switchport mode access
IND-ED-TEST(config-if)#switchport access vlan 40
IND-ED-TEST(config-if)#exit
IND-ED-TEST(config)#interface GigabitEthernet1/2
IND-ED-TEST(config-if)#switchport mode access
IND-ED-TEST(config-if)#switchport access vlan 50
IND-ED-TEST(config-if)#exit

```

## Where we at son?

```
IND-ED-TEST(config)#do show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa1/1
10	USERS	active	Fa1/2
20	VOICE	active	Fa1/3
30	MGMT	active	Fa1/4
40	IOT	active	Gi1/1
50	GUEST	active	Gi1/2
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

```
IND-ED-TEST(config)#
```

## 4. Give VLAN MGMT an IP Address - 192.168.10.2

```

IND-ED-TEST#
IND-ED-TEST#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
IND-ED-TEST(config)#interface vlan 30
IND-ED-TEST(config-if)#
*Mar 1 17:21:20.070: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30,
changed state to down
IND-ED-TEST(config-if)#ip address 192.168.10.2 255.255.255.0
IND-ED-TEST(config-if)#no shutdown
IND-ED-TEST(config-if)#exit
IND-ED-TEST(config)#ip default-gateway 192.168.10.1
IND-ED-TEST(config)#do show ip interface brief
Interface IP-Address OK? Method Status Protocol
Vlan1 unassigned YES unset administratively down down
Vlan30 192.168.10.2 YES manual up down
FastEthernet1/1 unassigned YES unset down down
FastEthernet1/2 unassigned YES unset down down
FastEthernet1/3 unassigned YES unset down down
FastEthernet1/4 unassigned YES unset down down
GigabitEthernet1/1 unassigned YES unset down down
GigabitEthernet1/2 unassigned YES unset down down
IND-ED-TEST(config)#

```

 **NOTE** - In addition to setting the IP address of the switch, you should give it a logical hostname and domain name in order to enable SSH. SSH requires a unique device hostname and an active domain name to generate encryption keys. You also need to configure a local administrator username/password and a privileged enable secret

```

IND-ED-TEST(config)#crypto key generate rsa modulus 2048
% Please define a domain-name first.
IND-ED-TEST(config)#

```

## 5. *Enable SSH and Telnet*

 **NOTE** - Telnet requires a privileged EXEC (enable) password or secret on older IOS versions before permitting login access:

```
IND-ED-TEST(config)#username Ind-it secret [insert password]
IND-ED-TEST(config)#enable secret [insert enable password]
IND-ED-TEST(config)#ip domain-name hulamin.co.za
IND-ED-TEST(config)#crypto key generate rsa modulus 2048
The name for the keys will be: IND-ED-TEST.hulamin.co.za

% The key modulus size is 2048 bits
% Generating 2048 bit RSA keys, keys will be non-exportable...
[OK] (elapsed time was 26 seconds)

IND-ED-TEST(config)#
*Mar 1 17:38:05.864: %SSH-5-ENABLED: SSH 1.99 has been enabled
IND-ED-TEST(config)#ip ssh version 2
IND-ED-TEST(config)#ip ssh time-out 60
IND-ED-TEST(config)#line vty 0 15
IND-ED-TEST(config-line)#transport input ssh telnet
IND-ED-TEST(config-line)#login local
IND-ED-TEST(config-line)#exit
IND-ED-TEST(config)#exit
IND-ED-TEST#
IND-ED-TEST#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
IND-ED-TEST#
```

## Test SSH/Telnet Connectivity

I'm sure you noticed at the very beginning when I listed everything you need for this task, from that list we didn't include a router but the very first thing we did include was **a LAN cable**. In order to wirelessly connect to a Switch via SSH/Telnet we need to establish a connection (a network).

 **NOTE - Some cisco routers do have built-in Wi-Fi, but most enterprise and industrial routers rely on separate access points.**

You may assume you've already created a network by connecting the PC to the Switch via Serial-to-USB but a PC connected to a network switch using a serial-to-USB console cable is **NOT a network connection**. You cannot assign an IP address to a physical console port because it is a serial or asynchronous management port, not a network interface. Instead, you use the console port via a physical cable to log into the device and assign an IP address to a network interface (like an Ethernet port or Switch Virtual Interface/VLAN).

To connect your personal PC directly to the switch without an external network or router, you need to create a simple direct subnet between your personal PC's Ethernet port and an switch Ethernet port. To create a simple basic network, we connect a PC to

a Switch via LAN cable using the port we assigned an IP address to on the switch. Do the following:

1. You should have set up the Switch Management IP above, so all we need to do is change IP settings on your personal PC/laptop (PC). Because there is no network or DHCP server to assign an IP address to your personal PC, you must set one manually:
  - **Windows:** Go to **Settings > Network & Internet > Ethernet > Edit IP assignment to Manual (IPv4)**.
  - **IP Address:** **192.168.1.2**
  - **Subnet Mask:** **255.255.255.0**
  - **Gateway / DNS:** Leave blank.
2. Run an Ethernet cable directly from your personal PC's network port into **FastEthernet1/4 (VLAN 30)** on the switch. You should see this on your terminal:

```
IND-ED-TEST>
*Mar 1 19:04:05.663: %LINK-3-UPDOWN: Interface FastEthernet1/4, changed state to
up
*Mar 1 19:04:06.670: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet1/4, changed state to up
*Mar 1 19:04:33.681: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30,
changed state to up
IND-ED-TEST>
```

3. Ping your Switch from your computer and vice versa on your switch

```
IND-ED-TEST>ping 192.168.10.3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.3, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/5/9 ms
```

 **NOTE - If you try and ping the PC you are connected via Serial-to-USB with, you will be ghosted because these two aren't connected on a network son.**

```
IND-ED-TEST>ping 10.10.22.63
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.22.63, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
IND-ED-TEST>
```

For a complete step-by-step walkthrough of SSH commands and key generation on Cisco switches, watch [SSH Configuration on Cisco Routers and Switches](#). This video provides a clear visual demonstration of setting up RSA keys, user accounts, and testing SSH connections using terminal software.

## One final step...

Okay look over here...



## What do you remember?

Nothing? Cool okay, so let's start over buddy!!!

Since you don't remember anything let's reload the switch so we can start from the beginning with a fresh unconfigured switch.

1. *First we have to delete the VLAN Database (vlan.dat)*



```

IND-ED-TEST>
IND-ED-TEST>enable
Password:
IND-ED-TEST#show flash:

Directory of flash:/

 2 -rwx 1581 Mar 1 1993 02:33:09 +00:00 starup-config
 3 drwx 1024 Mar 1 1993 00:00:22 +00:00 ies-lanbasek9-mz.152-4.EA5
 648 -rwx 38588 Mar 1 1993 00:01:04 +00:00 pnap.dat
 649 -rwx 856 Mar 1 1993 17:00:13 +00:00 vlan.dat
 650 -rwx 1314 Mar 1 1993 00:27:49 +00:00 start-config
 652 -rwx 1541 Mar 1 1993 17:53:12 +00:00 config.text
 653 -rwx 3576 Mar 1 1993 17:53:12 +00:00 private-config.text
 654 -rwx 6168 Mar 1 1993 17:53:12 +00:00 multiple-fs

64094208 bytes total (39586816 bytes free)
IND-ED-TEST#delete flash:vlan.dat
Delete filename [vlan.dat]?
Delete flash:/vlan.dat? [confirm]
IND-ED-TEST#

```

## 2. *Now delete(erase) the startup-config*

```

IND-ED-TEST#
IND-ED-TEST#erase startup-config
Erasing the nvram filesystem will remove all configuration files! Continue?
[confirm]
[OK]
Erase of nvram: complete
IND-ED-TEST#
*Mar 1 22:33:43.516: %SYS-7-NV_BLOCK_INIT: Initialized the geometry of nvram
IND-ED-TEST#

```

## 3. *Now let's reload the switch*

```

IND-ED-TEST#reload
Proceed with reload? [confirm]

*Mar 1 22:37:12.292: %SYS-5-RELOAD: Reload requested by console. Reload Reason:
Reload command.

```

- When it's done reloading the switch will ask you to **press RETURN to get started!**, press Enter and in the System Configuration Dialog enter **no**, aaaaaaand you're done, you now have a naked switch!

Press RETURN to get started!

```
*Mar 1 00:00:18.975: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1,
changed state to down
*Mar 1 00:00:20.501: %SPANTREE-5-EXTENDED_SYSID: Extended SysId enabled for type
vlan
*Mar 1 00:00:46.238: %PHY-5-TRANSCIEVERINSERTED: Slot=1 Port=1: Transceiver has
been inserted
*Mar 1 00:00:47.563: %SYS-5-RESTART: System restarted --
Cisco IOS Software, IES Software (IES-LANBASEK9-M), Version 15.2(4)EA5, RELEASE
SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2016 by Cisco Systems, Inc.
Compiled Tue 20-Dec-16 14:59 by prod_rel_team
*Mar 1 00:00:48.268: %SYS-5-CONFIG_I: Configured from console by console
```

--- System Configuration Dialog ---

Enable secret warning

-----

In order to access the device manager, an enable secret is required

If you enter the initial configuration dialog, you will be prompted for the enable secret

If you choose not to enter the initial configuration dialog, or if you exit setup without setting the enable secret,

please set an enable secret using the following CLI in configuration mode-  
enable secret 0 <cleartext password>

-----

Would you like to enter the initial configuration dialog? [yes/no]: no  
Switch>

Now go back to the tippity-top like a good lil boy and **DO IT AGAIN!**